INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALT LAKE
(Affiliated to UEM Kolkata and Approved by AICTE)

Degree: BCA, Stream: BCA

Term – I , Third Semester Examination, September - 2024

Subject Code: BCACC301

Subject Name: Data Structures with C

Full Marks: 30 **Duration: 1 Hour**

Part - A
Attempt 5 questions
Each question carries 2 marks (2 × 5)

1. Describe in brief the various data structures. 2

or

Define the FIFO principal & How it is different from LIFO? 2

2. Describe dynamic memory allocation? 2

or

Differentiate between primitive and non-primitive data structures? 2

3. Define Queue. 2

or

What does 'front' signifies in queue. 2

4. Define the top pointer of a stack. 2

or

Interpret the condition to check a stack is full or not. 2

5. Describe the time complexity of Binary search. 2

or

Differentiate between Malloc() and Realloc() functions 2

Part - B
Attempt 2 questions
Each question carries 5 marks (5 × 2)

6. Examine Deque in data structure? Write a C function to insert elements in linear queue. 5

or

Demonstrate the algorithms for the insertion and deletion operations performed on circular queue 5

7. Solve the expression from infix to postfix using stack 5
(a - b) / (c ^ d) + (e * f)

or

Illustrate a C function to display the contents of pointers using an array of pointers **5**

Part - C

Attempt 1 question

Each question carries 10 marks (10 × 1)

8. Design a C program to add two matrixes and display the summation. **10**

or

Design a C program to implement circular queue for insertion and deletion of elements in the queue. **10**

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